

VDI Infrastructure - Design & Planning

Centralizing Desktop Workplaces Across a Multi-Site Enterprise

Project Type	Network & Infrastructure / VDI Architecture
Role	IT Architect & Network Planner
Year	2026
Tech	Cisco Meraki SD-WAN, VDI-Server, Cisco AnyConnect VPN, Cisco ASA
Sites	11 Corporate Locations + Remote/Home Office (Germany)
Scope	Proof of Concept (25 users) → Full Production Rollout (~100 users)

1.1 SITUATION & PROBLEM STATEMENT

The company operated decentralized IT workplaces across eleven German locations (Zwickau, Berlin, Dresden, Leipzig, Potsdam, Mainz, Bochum, Lauta, Freiberg, Halle, Cottbus, Gera, Chemnitz), with IT headquarters in Zwickau. Each site managed its own client hardware locally with no centralized desktop management in place.

Key Challenges

- High overhead for decentralized hardware maintenance and support across all sites
- No cross-site collaboration platform or unified user experience
- Expensive, heterogeneous client hardware required at every location
- Time-consuming onboarding for new staff or new office openings

The existing network used Cisco Meraki MX appliances as SD-WAN gateways at all sites. Zwickau and Berlin featured MX-100 units with EC 2.0-200 / DCIP-1GB (1 Gbit/s). Remaining sites ran MX-68 with DCIP-50 or DCIP-100. Meraki Mesh served as a secondary backup link.

1.2 PROJECT OBJECTIVES

The goal was to design a centralized VDI with a phased rollout, establishing the primary VDI site in Zwickau (HQ):

- Flexible workspace delivery - staff can work from any site or home office
- Cost reduction - no expensive local client hardware; a standard laptop suffices
- Rapid onboarding - new offices or home setups connectable without hardware investment
- Centralized desktop management - single platform for all 11+ sites
- High availability - redundant operation with failover to Berlin or an external data center

1.3 PROJECT APPROACH & PHASES

Phase	Name	Scope	Key Detail
Phase 1	Pre-VDI	All 11 sites, decentralized	Local clients, no central management
Phase 2	Proof of Concept	4 pilot sites (~25-40 users)	VDI server in Zwickau; external DC evaluated
Phase 3	Full Production Rollout	All 11 sites (~100 users)	Failover: Berlin or external data center

Phase 1 - Analysis

Comprehensive audit and documentation of the existing network infrastructure. All sites operated local clients with on-premise hardware. Home office users accessed corporate resources via Cisco AnyConnect VPN (192.168.80.0/24). No centralized desktop management was in place.

Phase 2 - Proof of Concept

Four sites selected for the PoC: Bochum, Lauta, Mainz, and Potsdam (~25 users, expandable to ~40). The VDI server was deployed in Zwickau; Q.Beyond external data center was evaluated as an alternative. Bandwidth assessments confirmed sufficient capacity at three of four sites (Lauta flagged for monitoring at 7 users on a 50 Mbit line).

Phase 3 - Full Production Rollout

- Option A: Primary VDI in Zwickau with Berlin as active failover (identical bandwidth capacity)
- Option B: Full operation in external data center (Cisco ASA + VDI server + all services)
- Home office: Cisco AnyConnect VPN via Meraki Z3 gateway retained; desktop delivered via VDI

2. TECHNICAL IMPLEMENTATION

2.1 Network Architecture & SD-WAN

The corporate network is built on Cisco Meraki SD-WAN. Each site connects via a primary DSL link with a Meraki Mesh backup. MX-68 units automatically fail over to Mesh if the primary circuit drops. Zwickau and Berlin anchor the architecture with 1 Gbit/s symmetrical connectivity; branch sites operate on 50-100 Mbit links.

Site overview: Zwickau (MX-100, 1 Gbit/s, Primary VDI Host) | Berlin (MX-100, 1 Gbit/s, Failover Site) | Dresden, Leipzig, Freiberg, Cottbus (MX-68, 100 Mbit, Branch) | Bochum, Lauta, Mainz, Halle, Gera (MX-68, 50 Mbit, Branch) | Potsdam (MX-68, 100 Mbit, PoC Site)

2.2 VDI Infrastructure

- Primary VDI host in Zwickau: Cisco Meraki MX-100, DCIP-1GB sufficient capacity for full workload
- Failover: VDI Backup Server in Berlin with identical bandwidth capacity
- External DC (Q.Beyond): Cisco ASA firewall; VDI server + NetPhone, Web/Email Secure, Exchange
- Home office: Cisco Meraki Z3 as VPN gateway, Cisco AnyConnect VPN (192.168.80.0/24)
- Meraki Mesh exception routing: specific devices route via defined paths (e.g., Potsdam to Berlin)

3. RESULTS & BUSINESS VALUE

- Complete network documentation
- PoC concept - detailed plan for 4 pilot sites with bandwidth analysis and go/no-go assessment
- Scaling roadmap - phased rollout plan to full production across all locations
- Redundancy concept - three operational scenarios: Zwickau-only, Dual-Site Failover, External DC
- Executive decision package - network diagrams and scenario comparisons for management sign-off

Quantified Impact

- Eliminated per-site client hardware investment - standard laptops sufficient across all locations
- Onboarding time for new sites reduced from weeks to days
- Centralized management for ~100 users across 11+ sites from a single platform
- Seamless home office integration - identical desktop experience to office workstations
- High availability through Dual-Site operation (Zwickau + Berlin) or external DC failover

4. SUMMARY

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Architected a centralized VDI for an 11-site, ~100-user enterprise in Germany — analyzing the existing Cisco Meraki SD-WAN environment, designing a phased PoC-to-production rollout, and delivering three redundancy scenarios (on-premises, dual-site failover, external DC) with full network documentation and executive decision packages.

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